环境

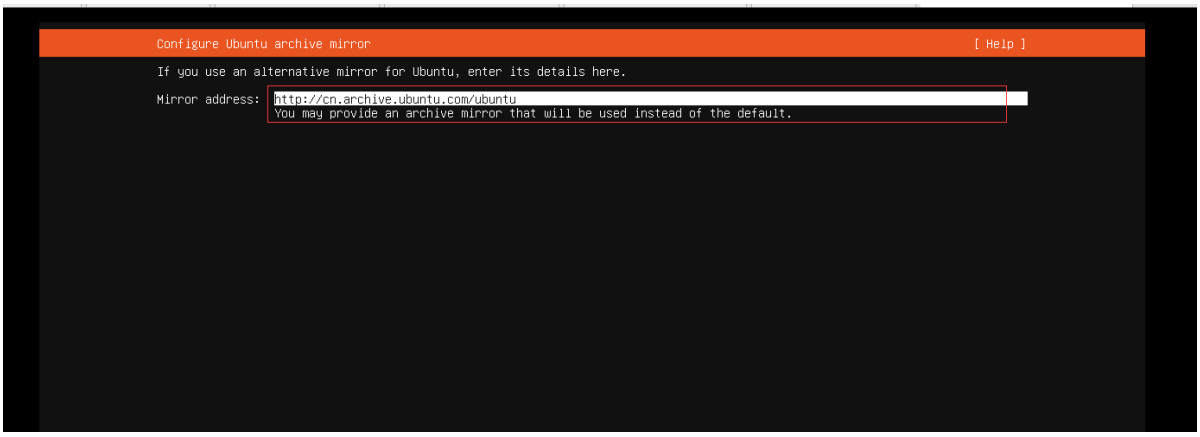
主机环境

1. 宿主机环境 ubuntu-20.04.4-live-server-amd64，下载地址：

https://mirrors.aliyun.com/ubuntu-releases/20.04.4/ubuntu-20.04.4-live-server-amd64.iso

2. apt 包管理器，镜像源修改:

将 http://cn.archive.ubuntu.com/ubuntu 改为 https://mirrors.tuna.tsinghua.edu.cn/ubuntu
两种修改方式：
第一种：在安装虚拟机时，修改图下图：



第二种：已经安装好系统的情况下，修改 /etc/apt/sources.list 将对应的地址替换，替换完成后执行 apt-get update。注意：先备份/etc/apt/sourse.list文件

docker 安装

基于apt包管理器安装

1. 安装

```
sudo apt  install docker.io
```

2. 卸载

```
sudo apt-get purge docker.io
sudo rm -rf /var/lib/docker
sudo rm -rf /var/lib/containerd
```

根据官方文档安装

1. 官方文档：https://docs.docker.com/engine/install/ubuntu/
2. 有三种安装方式：1. 基于官方存储库安装；2. 下载软件包安装；3. 基于官方给出的快捷脚本安装。

使用docker官方存储库安装

1. 更新 apt 包索引，并且安装一些软件使得apt可以通过HTTPS协议访问软件库。

```
 sudo apt-get update
 sudo apt-get install \
    ca-certificates \
    curl \
    gnupg \
    lsb-release
```

2. 添加docker官方的GPG 秘钥

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/docker-archive-keyring.gpg
```

3. 设置一个标准的docker软件仓库。

```
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-
archive-keyring.gpg] https://download.docker.com/linux/ubuntu \
  $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list >
/dev/null
```

4. 更新apt 包索引并查看docker 可用版本列表

```
sudo apt-get update
apt-cache madison docker-ce
```

5. 安装特定版本docker语法，例如：5:24.0.2-1~ubuntu.18.04~bionic

```
sudo apt-get install docker-ce=<VERSION_STRING> \
docker-ce-cli=<VERSION_STRING> \
containerd.io=<VERSION_STRING> \
docker-compose-plugin=<VERSION_STRING>
```

```
sudo apt-get install docker-ce=5:24.0.5-1~ubuntu.20.04~focal \
docker-ce-cli=5:24.0.5-1~ubuntu.20.04~focal \
containerd.io=1.6.21-1 \
docker-compose-plugin=2.20.2-1~ubuntu.20.04~focal
```

6. 安装最新版docker-ce及相关软件

```
sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

7. 检查docker engine是否正确安装

```
docker run hello-world
```

8. 检查服务

```
nick@master1:~$ systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
     Active: active (running) since Fri 2023-08-04 11:53:19 UTC; 59s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 4655 (dockerd)
      Tasks: 12
     Memory: 29.9M
     CGroup: /system.slice/docker.service
             └─4655 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.538147864Z" level=info msg="Starting up"
Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.540679731Z" level=info msg="detected 127.0.0.53 nameserver, assuming systemd-resolved, so using resolv.conf: /ru
Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.594995143Z" level=info msg="Loading containers: start."
Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.778002452Z" level=info msg="Default bridge (docker0) is assigned with an IP address 172.17.0.0/16. Daemon option
Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.873501954Z" level=info msg="Loading containers: done."
Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.912637593Z" level=warning msg="WARNING: No swap limit support"
Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.912745549Z" level=info msg="Docker daemon" commit=a61e2b4 graphdriver=overlay2 version=24.0.5
Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.913085983Z" level=info msg="Daemon has completed initialization"
Aug 04 11:53:19 master1 dockerd[4655]: time="2023-08-04T11:53:19.979020740Z" level=info msg="API listen on /run/docker.sock"
Aug 04 11:53:19 master1 systemd[1]: Started Docker Application Container Engine.
lines 1-21/21 (END)
```

卸载

1. 卸载

```
sudo apt-get purge docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

2. 删除相关目录

```
sudo rm -rf /var/lib/docker && sudo rm -rf /var/lib/containerd && sudo rm -rf
/etc/docker/
```

将用户添加到docker组

将用户添加到docker用户组后，不需要每次都输入sudo来执行docker命令了

```
//将用户从docker用户组中移除 gpasswd -d <username> docker
//将用户添加到docker 用户组
sudo addgroup -a <username> docker
sudo service docker restart
//查看用户信息
id <username>
```

退出终端，重新连接即可